

Question: 1

Given the network 192.168.10.0/24, divide it into 4 equal subnets and write down the network address, first usable IP, last usable IP, and broadcast address for each subnet.

Answer: 1

Subnet Division Strategy

To divide the base 192.168.10.0/24 network into 4 equal subnets, we take the total pool of 256 addresses and divide them by 4, giving 64 addresses per block (256 ÷ 4 = 64).

A block size of 64 requires borrowing 2 bits from the host portion, which changes the subnet prefix from /24 to a /26 prefix (Subnet Mask: 255.255.255.192).

Equal Subnet Allocation Table

Subnet Number	Network Address	First Usable IP	Last Usable IP	Broadcast Address	Total Usable Hosts
Subnet 1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63	62 Hosts
Subnet 2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127	62 Hosts
Subnet 3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191	62 Hosts
Subnet 4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255	62 Hosts

Question: 2

You are setting up a Zomato-style food delivery office with 3 departments: Customer Support (50 devices), Delivery Management (30 devices), and Marketing (20 devices). Design a subnetting plan using the 10.0.0.0/24 network so that each department has its own subnet, maximizing available addresses and minimizing waste. List the subnet mask and IP range for each department.

Answer: 2

VLSM Department Subnet Allocation Table

To design an efficient plan using **Variable Length Subnet Masking (VLSM)**, the departments are sorted from the **largest to the smallest device requirement** to eliminate IP address wastage:

Department Hierarchy	Device Count	Subnet Block Size	CIDR Notation	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address
1. Customer Support	50 Devices	64 IPs	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62	10.0.0.63
2. Delivery Management	30 Devices	32 IPs	10.0.0.64/27	255.255.255.224	10.0.0.65	10.0.0.94	10.0.0.95
3. Marketing	20 Devices	32 IPs	10.0.0.96/27	255.255.255.224	10.0.0.97	10.0.0.126	10.0.0.127

Quick Setup Logic

- Customer Support (50 devices):** The smallest power of 2 that holds 50 devices is $2^6 = 64$ absolute IPs ($64 - 2 = 62$ usable). This requires a **/26** prefix, using up slots **.0** to **.63**.
- Delivery Management (30 devices):** The smallest power of 2 that holds 30 devices is $2^5 = 32$ absolute IPs ($32 - 2 = 30$ usable). This fits perfectly into a **/27** prefix, using up slots **.64** to **.95**.
- Marketing (20 devices):** The smallest power of 2 that holds 20 devices is also $2^5 = 32$ absolute IPs ($32 - 2 = 30$ usable). This uses a **/27** prefix, occupying slots **.96** to **.127**.

Question: 3

Troubleshoot the following scenario: Two teams in your Flipkart warehouse are accidentally assigned overlapping subnets (Team A: 172.16.10.0/25, Team B: 172.16.10.64/26). Identify the overlapping IP range and suggest a corrected subnet plan so both teams have unique IP ranges.
Hint: Draw the address ranges for both subnets to spot the overlap.

Answer: 3

1. Identification of the Overlapping IP Range

To locate the conflict, we break down the absolute boundary ranges for both assigned network configurations:

- **Team A Allocation (172.16.10.0/25):**
 - Total block size = $2^7 = 128$ absolute IP addresses.
 - Complete Range: 172.16.10.0 to 172.16.10.127
- **Team B Allocation (172.16.10.64/26):**
 - Total block size = $2^6 = 64$ absolute IP addresses.
 - Complete Range: 172.16.10.64 to 172.16.10.127
- **The Conflict:** 172.16.10.64 to 172.16.10.127 is the overlapping range. Team B's entire subnetwork sits inside the upper half of Team A's larger subnet pool, which will trigger immediate IP duplication errors and connection drops on the warehouse floor.

2. Corrected Subnet Plan

To fix the overlap while maintaining the exact team block sizes (128 addresses for Team A and 64 addresses for Team B), Team B is shifted completely past Team A's boundary marker:

Warehouse Allocation	Corrected CIDR Block	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address
Team A (Unchanged)	172.16.10.0/25	255.255.255.128	172.16.10.1	172.16.10.126	172.16.10.127
Team B (Shifted)	172.16.10.128/26	255.255.255.192	172.16.10.129	172.16.10.190	172.16.10.191

3. Why this Fix Works

By relocating Team B to start exactly at the next sequential block boundary (**.128**), both warehouse sectors now operate on completely separated, unique IP subnets. This resolves all network address space collisions without altering Team A's active assignments.

Question: 4

You are managing a growing IPL Fantasy League app backend and notice that your 192.168.1.0/26 subnet is running out of IPs. Propose a solution to handle address exhaustion so new servers can be added without disrupting existing users.

Hint: Consider subnet resizing or migrating to a larger subnet.

Answer: 4

Proposed Solution: Subnet Expansion via Supernetting

The current subnet is **192.168.1.0/26**, which only provides a total of **62 usable IP addresses** (192.168.1.1 to 192.168.1.62). Because this pool is running out of IPs, new fantasy league servers cannot connect.

To resolve this address exhaustion without disrupting the active, existing servers, the best approach is to expand the current subnet mask by changing the prefix from /26 to a **/24 network configuration**.

Before & After Subnet Transition Table

Network Parameter	Old Limited Subnet	New Expanded Subnet	Operational Impact
CIDR Notation	192.168.1.0/26	192.168.1.0/24	Keeps the same base network footprint.
Subnet Mask	255.255.255.192	255.255.255.0	Vertically opens up the host address pool bits.
First Usable IP	192.168.1.1	192.168.1.1	No disruption ; core gateway router IP stays identical.
Last Usable IP	192.168.1.62	192.168.1.254	Extends the upper boundary threshold significantly.
Total Usable IPs	62 Hosts	254 Hosts	Instantly provides 192 new server slots .

Why this Approach is Efficient

- **Zero Disruption to Existing Servers:** The existing servers (running on IPs .1 through .62) do not need to change their IP addresses, shut down, or migrate. Their current configurations sit perfectly inside the lower quadrant of the new /24 space.
- **Instant Configuration Update:** The system administrator only needs to update the Subnet Mask value from 255.255.255.192 to 255.255.255.0 on the central router and the main server pool.
- **Eliminates Migration Overhead:** Instead of building a completely new network block and moving data, widening the network boundary in place provides an immediate growth runway for new backend database instances.

Question: 5

After completing your subnetting speed round and mock exam, review your answers and identify 3 types of mistakes you made (e.g., miscalculating subnet masks, overlapping ranges, wrong broadcast address). For each, write how you would avoid the mistake next time.

Answer: 5

3 Common Subnetting Mistakes & Prevention Strategies

- **Mistake 1: Miscalculating the Subnet Mask Value**
 - **How I Made It:** When working quickly under exam pressure, I accidentally wrote the wrong decimal number for tight subnets (like writing .224 instead of .240 for a /28 mask).
 - **How to Avoid Next Time:** I will write down a quick **8-bit reference line** (128, 192, 224, 240, 248, 252, 254, 255) at the very top of my rough sheet before starting the exam to look up values instantly instead of doing mental math every time.
- **Mistake 2: Overlapping Subnet IP Ranges**
 - **How I Made It:** When designing Variable Length Subnet Masking (VLSM) plans, I sometimes started the next department's subnet on a random number rather than aligning it strictly with the **next boundary block size multi-step**.
 - **How to Avoid Next Time:** I will strictly follow the rule of sorting my device requirements from the **largest department to the smallest department** first. I will always make sure that each new subnet address is perfectly divisible by its total block size before assigning it.
- **Mistake 3: Writing the Wrong Broadcast Address**
 - **How I Made It:** I occasionally forgot that the absolute first address (Network ID) is counted as **0**, which caused me to add the block size directly and overshoot the broadcast limit by one number.

- **How to Avoid Next Time:** I will systematically use the simple subtraction checkpoint formula:

$$\text{Broadcast Address} = (\text{Network Address} + \text{Total Block Size}) - 1$$

Always subtracting one will ensure I never hit the next subnet's territory.